

CENTRAL BOARD OF SECONDARY EDUCATION**HALF-YEARLY PRACTICE EXAMINATION (SESSION 2024-2025)****SUBJECT: SCIENCE (CLASS - IX) | CHAPTER 5: EXPLORING MIXTURES AND THEIR SEPARATION****TIME ALLOWED: 45 MINUTES****MAXIMUM MARKS: 25****PAPER CODE: 086/CH05****Student's Name:** _____**Roll No.:** _____**Class & Section:** Class IX - _____**Date of Exam:** _____**Invigilator's Sign:** _____**GENERAL INSTRUCTIONS:**

1. The question paper comprises **25 Multiple Choice Questions (MCQs)** of **1 mark each**.
2. **Section A (Q1 – Q15):** Standard Multiple Choice Questions with one correct option.
3. **Section B (Q16 – Q20):** Assertion-Reasoning based questions. Read both statements carefully before marking.
4. **Section C (Q21 – Q25):** Case-Study / Practical Data based questions.
5. All questions are compulsory. There is no negative marking.
6. Darken the corresponding circle on the **OMR Sheet** using a blue or black ballpoint pen.

SECTION A: MULTIPLE CHOICE QUESTIONS (Q.1 TO Q.15)**[15 MARKS]****Q.1 Which of the following is a homogeneous mixture?**

- (A) Chalk powder in water (B) A brass alloy of copper and zinc
(C) Muddy pond water (D) Oil and water mixture

Q.2 What is the mass by mass percentage concentration of a solution prepared by dissolving 40 g of common salt in 360 g of water?

- (A) 11.1% (B) 10.0%
(C) 9.0% (D) 12.5%

Q.3 Which property correctly characterizes a true solution?

- (A) Particle size is greater than 100 nm (B) Solute particles settle down on standing
(C) Particles pass completely through standard filter paper (D) It clearly scatters a light beam (Tyndall effect)

Q.4 The scattering of a visible beam of light when passed through a colloidal solution is known as:

- (A) Raman effect (B) Tyndall effect
(C) Doppler effect (D) Photoelectric effect

Q.5 Which of the following mixtures will show a noticeable Tyndall effect?

- (A) Salt solution (B) Copper sulphate solution
(C) Milk (D) Sugar solution

Q.6 In a colloidal aerosol like fog, clouds, and mist, what are the dispersed phase and dispersion medium?

- (A) Dispersed phase: Liquid, Dispersion medium: Gas (B) Dispersed phase: Gas, Dispersion medium: Liquid
(C) Dispersed phase: Solid, Dispersion medium: Liquid (D) Dispersed phase: Gas, Dispersion medium: Gas

Q.7 Which separation technique is based on difference in density where heavier particles are forced to the bottom by rapid spinning?

- | | |
|-----------------------------|--------------------|
| (A) Fractional distillation | (B) Centrifugation |
| (C) Sublimation | (D) Chromatography |

Q.8 Two miscible liquids having boiling points of 65°C and 78°C (difference of 13°C) can be best separated using:

- | | |
|-------------------------|-----------------------------|
| (A) Simple distillation | (B) Fractional distillation |
| (C) Separating funnel | (D) Filtration |

Q.9 Which mixture can be separated by the method of sublimation?

- | | |
|-------------------------------|---|
| (A) Sodium chloride and water | (B) Ammonium chloride and sodium chloride |
| (C) Iron filings and sand | (D) Water and kerosene oil |

Q.10 The technique of paper chromatography separates solutes present in a mixture based on differences in their:

- | | |
|--|-------------------------|
| (A) Densities | (B) Boiling points |
| (C) Solubility in the mobile solvent phase | (D) Magnetic properties |

Q.11 Crystallization is preferred over simple evaporation to dryness for purifying salts like copper sulphate because:

- | | |
|---|---|
| (A) It requires less time | (B) Some solids decompose or get charred upon heating to complete dryness |
| (C) Evaporation removes all impurities completely | (D) Crystallization requires toxic solvents |

Q.12 A mixture of kerosene oil and water is heterogeneous and can be separated using:

- | | |
|---------------------------|--------------------------|
| (A) Fractionating column | (B) A separating funnel |
| (C) Sublimation apparatus | (D) Chromatography paper |

Q.13 What type of colloid is shaving cream, where gas is dispersed in a liquid?

- | | |
|----------|--------------|
| (A) Sol | (B) Gel |
| (C) Foam | (D) Emulsion |

Q.14 When a beam of sunlight enters a dusty, dimly lit room through a small window hole, the path of light becomes visible because:

- | | |
|---|---|
| (A) Light is absorbed by dust | (B) Air dust and smoke particles scatter light (Tyndall effect) |
| (C) Sunlight consists of ultraviolet rays | (D) Light slows down to zero speed |

Q.15 A solution that contains the maximum amount of solute dissolved at a given specific temperature is termed a:

- | | |
|-------------------------------|------------------------|
| (A) Unsaturated solution | (B) Saturated solution |
| (C) Supersaturated suspension | (D) Colloid |

SECTION B: ASSERTION - REASON QUESTIONS (Q.16 TO Q.20)

[5 MARKS]

Directions for Q16 to Q20: Select the correct option from the choices given below:

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

Q.16 Assertion (A): Air is considered a homogeneous mixture of gases.

Reason (R): The gaseous components of air are uniformly mixed and cannot be distinguished by physical boundaries.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

Q.17 Assertion (A): A suspension is a heterogeneous mixture and its particles settle down under gravity when left undisturbed.

Reason (R): The solute particles in a suspension are larger than 100 nm and are visible to the naked eye.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

Q.18 Assertion (A): Sugar dissolved in water can be separated by filtration using filter paper.

Reason (R): Sugar solution is a true solution whose particle size is less than 1 nm.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

Q.19 Assertion (A): Fractional distillation is used to separate different gases from liquid air.

Reason (R): Different gases in air have different boiling points with differences greater than 100°C.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

Q.20 Assertion (A): Tincture of iodine is a solution of solid iodine dissolved in liquid alcohol.

Reason (R): In tincture of iodine, alcohol acts as the solute and iodine acts as the solvent.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

SECTION C: CASE-BASED / EXPERIMENTAL QUESTIONS (Q.21 TO Q.25)

[5 MARKS]

READING PASSAGE & EXPERIMENTAL SETUP:

Case Study: Forensic Ink Analysis by Chromatography

A forensic examiner is investigating a suspicious cheque where the written payment amount seems to have been altered with a second black ink pen. The examiner places a drop of the cheque's ink alongside standard ink samples from Pen X and Pen Y onto a strip of chromatography filter paper. The paper is suspended in a beaker containing a solvent of water and ethanol. As the solvent rises, the cheque ink splits into three distinct colored dye spots (blue, yellow, pink) that match exactly with the spot heights of Pen Y, while Pen X yields only blue and purple spots.

Q.21 What is the physical principle underlying the separation in paper chromatography?

- (A) Differential boiling points of dyes
(B) Differential solubility and adsorption rates of pigments in the mobile solvent
(C) Difference in magnetic attraction
(D) Sublimation temperatures of ink

Q.22 Based on the examiner's observation, which pen was used to write or alter the cheque?

- (A) Pen X
(B) Pen Y
(C) Neither Pen X nor Pen Y
(D) Both pens simultaneously

Q.23 Which component in paper chromatography acts as the stationary phase?

- (A) The rising water-ethanol solvent (B) The strip of chromatography paper
(C) The ink solvent vapor (D) The beaker glass wall

Q.24 The dye component that travels the fastest and reaches highest on the paper strip is the one that:

- (A) Has the greatest mass (B) Is most soluble in the moving solvent
(C) Has the highest boiling point (D) Is least soluble in water

Q.25 Which of the following is another practical real-world application of chromatography?

- (A) Purifying drinking water from muddy lakes (B) Separating drugs from blood samples in pathology
(C) Extracting common salt from seawater (D) Separating iron scrap from garbage

CBSE CANDIDATE OMR ANSWER RESPONSE GRID

01 (A) (B) (C) (D)	02 (A) (B) (C) (D)	03 (A) (B) (C) (D)	04 (A) (B) (C) (D)	05 (A) (B) (C) (D)
06 (A) (B) (C) (D)	07 (A) (B) (C) (D)	08 (A) (B) (C) (D)	09 (A) (B) (C) (D)	10 (A) (B) (C) (D)
11 (A) (B) (C) (D)	12 (A) (B) (C) (D)	13 (A) (B) (C) (D)	14 (A) (B) (C) (D)	15 (A) (B) (C) (D)
16 (A) (B) (C) (D)	17 (A) (B) (C) (D)	18 (A) (B) (C) (D)	19 (A) (B) (C) (D)	20 (A) (B) (C) (D)
21 (A) (B) (C) (D)	22 (A) (B) (C) (D)	23 (A) (B) (C) (D)	24 (A) (B) (C) (D)	25 (A) (B) (C) (D)

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & DETAILED EXPLANATIONS

Q. No.	Correct	Conceptual Explanation & Syllabus Rationale
Q.1	(B)	Brass is a solid solution (alloy) having uniform composition and properties throughout, making it a homogeneous mixture.
Q.2	(B)	$\text{Mass of solution} = 40 + 360 = 400 \text{ g}$. $\text{Percentage} = \frac{40}{400} \times 100 = 10.0\%$.
Q.3	(C)	In a true solution, solute particles are $< 1 \text{ nm}$, pass through filter pores, do not scatter light, and never settle under gravity.
Q.4	(B)	Colloidal particles are large enough (1-100 nm) to scatter visible light rays, illuminating the path of the beam (Tyndall effect).
Q.5	(C)	Milk is a colloidal emulsion of fat and protein droplets in water, which scatters light and shows Tyndall effect.
Q.6	(A)	In fog and mist, tiny liquid water droplets (dispersed phase) are dispersed in air/gas (dispersion medium).
Q.7	(B)	Centrifugation spins mixtures at high speed, forcing denser particles to the bottom (used for separating cream from milk and blood diagnostics).
Q.8	(B)	When boiling point difference is less than 25 K (or 25°C), fractional distillation with a fractionating column is required.
Q.9	(B)	Ammonium chloride sublimates directly into vapor upon heating, leaving non-sublimable sodium chloride behind.
Q.10	(C)	Solutes with higher solubility travel faster with the solvent front up the chromatography paper, separating into distinct bands.
Q.11	(B)	Evaporating to dryness can decompose salts (charring) and leaves soluble impurities behind, whereas crystallization produces pure crystals.
Q.12	(B)	Immiscible liquids form distinct layers based on density and are separated using a separating funnel with a stopcock.
Q.13	(C)	A colloid with gas dispersed in a liquid medium is termed a foam.
Q.14	(B)	Suspended dust and smoke particles in air act as colloidal scattering centers, making the light beam visible.
Q.15	(B)	A saturated solution can dissolve no more solute at that specified temperature.
Q.16	(A)	Air has a uniform chemical composition of nitrogen, oxygen, argon, etc., without visible phase boundaries.
Q.17	(A)	Because suspension particles are heavy ($> 100 \text{ nm}$), thermal molecular motion cannot overcome gravity, causing sedimentation.
Q.18	(D)	Assertion is false (sugar particles pass through filter pores easily). Reason is true (true solution particle size $< 1 \text{ nm}$).
Q.19	(C)	Assertion is true. Reason is false: The boiling points of nitrogen (-196°C), argon (-186°C), and oxygen (-183°C) differ by only 3°C to 13°C (much less than 100°C).
Q.20	(C)	Assertion is true. Reason is false because iodine is the dissolved solid (solute) and alcohol is the dissolving liquid (solvent).
Q.21	(B)	Chromatography separates mixtures based on relative affinity for the stationary phase (paper) versus mobile phase (solvent).
Q.22	(B)	The dye components and spot positions of the cheque ink matched Pen Y identically.
Q.23	(B)	The porous paper matrix through which the solvent migrates is the stationary phase.
Q.24	(B)	Components with higher solubility in the solvent move faster and travel further along the paper.

Q. No.	Correct	Conceptual Explanation & Syllabus Rationale
Q.25		Medical and forensic laboratories use chromatography to isolate and identify pharmaceutical compounds and toxins from blood.